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Robust feature selection via simultaneous sapped norm and sparse regularizer minimization

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ABSTRACT

High dimension is one of the key characters of big data. Feature selection, as a framework to identify a small subset of illustrative and discriminative features, has been proved as a basic solution in dealing with high-dimensional data. In previous literatures, $\ell_{2,p}$-norm regularization was studied by many researches as an effective approach to select features across data sets with sparsity. However, $\ell_{2,p}$-norm loss function is just robust to noise but not considering the influence of outliers. In this paper, we propose a new robust and efficient feature selection method with emphasizing Simultaneous Capped ℓ_2-norm loss and $\ell_{2,p}$-norm regularizer Minimization (SCM). The capped ℓ_2-norm based loss function can effectively eliminate the influence of noise and outliers in regression and the $\ell_{2,p}$-norm regularization is used to select features across data sets with joint sparsity. An efficient approach is then introduced with proved convergence. Extensive experimental studies on synthetic and real-world datasets demonstrate the effectiveness of our method in comparison with other popular feature selection methods.

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1. Introduction

With the development of information technology, the dimensionality of the data used in data mining becomes more and more numerous. Such high dimensional data often contains thousands or millions of features such as face recognition [1], object recognition [2–5], emotion recognition [6], video semantic recognition [7–10] and bioinformatics [11–15]. The high dimension property prevent from developing efficient models to process the data. Furthermore, some features are irrelevant and redundant and the existence of these features may result in low efficiency, over-fitting and poor prediction performance in learning tasks [16–20]. And the high dimensionality can significantly alleviate the effect of the curse of dimensionality [16]. Hence, dimensionality reduction plays an important role of data preprocessing in many applications.

Feature selection and feature extraction are two main techniques for dimensionality reduction [1,21,22]. Feature extraction usually projects or maps the features into a lower dimensional and discriminative feature space. Other than feature exaction, feature selection identifies optimal subset from the original features and is preferred in many real applications, such as genetic analysis [12,13,23,24] and video semantic understanding [7,25]. It allows us to just focus on the feature subset that we are concerned about, but not the whole set. Additionally, the selected features by feature selection methods need to be collected or calculated, while all input features are required to obtain the low-dimensional representation in feature extraction methods. As a result, we focus on feature selection in this paper.

Feature selection, which is a process of selecting a subset of relevant features, is a key component in building machine learning models for further data processing such as classification, clustering and other tasks [21,26,27]. According to how the classification algorithm is incorporated in evaluating and selecting features, feature selection methods can be organized into three types in the literature: (1) filter methods [28–31] where the selection is independent of classifiers used, (2) wrapper methods [32,33] where a feature subset search algorithm is wrapped around the classification model and the feature subsets are scored based on their predictive power, and (3) embedded methods [34–36] which searches for an optimal subset of features in the process of classifier construction. Compared to wrapper methods and filter methods, embedded methods can ensure a good estimated performance in an efficient operation process in order to get an accurate calculation rapidly. As a result, in this paper we concentrate on the embedded methods for supervised feature selection.

Embedded feature selection methods have been developed widely in recent years. Devore and Peck proposed the student *t*-

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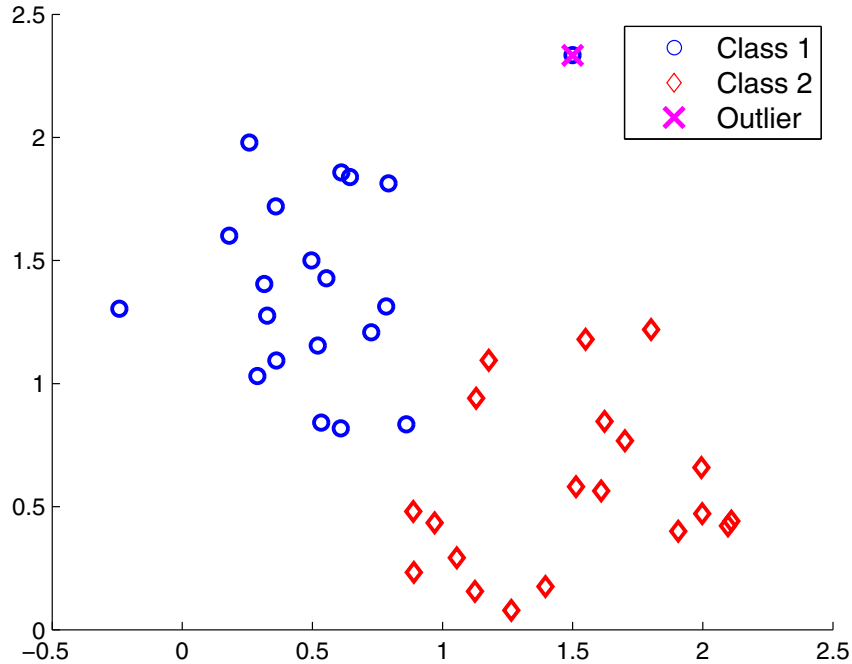


Fig. 1. The toy example.

test which can analysis the statistical discrepancy between two classes with regard to one certain feature [37]. Cover and Thomas put forward the theory of Informationgain based on the notion of entropy, which characterizes the impurity of an arbitrary set of examples [38]. Kira and Rendell raised the ReliefF algorithm, where features are selected according to their suitability for the target function [39]. Fisher score [40] is an effective selection method, which selects each feature independently according to their scores under the Fisher criterion. Particularly, Nie et al. proposed an Robust and efficient Feature Selection (RFS) emphasizing joint $\ell_{2,1}$-norm minimization on both loss function and regularization [13]. It is applied into both genomic and proteomic biomarkers discovery, and also outperforms other methods in many aspects. However, the $\ell_{2,1}$-norm loss is just robust to noise but not considering the influence of outliers. We demonstrate a 2D toy example adding an outlier in two classes of data sets as Fig. 1 shows. From an indepth study below, we find that RFS will project to y-axis when the outlier is existing. And when we remove the outlier, RFS will project to x-axis accurately instead. Therefore, the outliers can seriously influence the projection of RFS.

To deal with outliers more efficiently, we propose a new robust and efficient feature selection method emphasizing Simultaneous Capped ℓ_2-norm loss function and $\ell_{2,p}$-norm regularization Minimization (SCM). We use capped ℓ_2-norm based loss function to reduce the influence of outliers, and it is robust to noise as well [41]. Motivated by previous research [42–44], an $\ell_{2,p}$-norm regularization is performed to select features across all data points with sparsity when $0 < p \leq 1$. Compared with previous work in [41], there are several fundamental improvement of this paper: (1) we extend the $\ell_{2,1}$-norm regularization to $\ell_{2,p}$-norm, from which we can consider more general cases. Our previous method in [41] can be regarded as a special case of this work. Meanwhile, we analyze the parameters in detail and provide some experiments for demonstration. (2) We provide an efficient way to solve the general problem and prove the convergence in both theory and experiment. It is more complicated than that in [41]. (3) More comprehensive experimental results are provided to deepen the analyses of this problem. The contributions we make are summarized as,

- Propose a new formulation of robust and efficient feature selection method based on simultaneous capped ℓ_2-norm loss function and $\ell_{2,p}$-norm regularization minimization. The capped ℓ_2-norm loss is robust to both noise and outliers and the $\ell_{2,p}$-norm regularization is likely to offer better sparsity. Thus, the formulation is more specifically suited to feature selection.
- Develop an efficient algorithm to address the formulated minimization problem and rigorously analyze the performance of our method in aspects of the convergence behavior and parameter determination.
- Propose a 2D toy example to demonstrate that the capped ℓ_2-norm loss function is more robust than $\ell_{2,p}$-norm loss function especially in deposing of outliers.
- Evaluate SCM systematically on diverse kinds of real world data sets to demonstrate the performance of our method in both classification accuracy and redundancy rate comparing with five other feature selection methods.

The rest of this paper is organized as follows. Section 2 provides notations and related work. We formulate the proposed SCM algorithm and provide an effective approach to the formulated problem in Section 3. In Section 4, we strictly analyse the convergence behavior and parameter determination. Experimental results on various types of data sets are displayed in Section 5. Finally, we summarizes the paper in Section 6.

2. Related work

2.1. Notations and definitions

In this paper, matrices and vectors are written as boldface uppercase letters and boldface lowercase letters, respectively. For example, a matrix $\mathbf{M} \in \mathbb{R}^{s\times t}$, its ith row, jth column are denoted by $\mathbf{m}^i$, $\mathbf{m}_j$, respectively. The ℓ_p-norm of the vector $\mathbf{v} \in \mathbb{R}^n$ is defined as $\|\mathbf{v}\|_p = (\sum_{i=1}^n |v_i|^p)^{\frac{1}{p}}$. The ℓ_0-norm of the vector $\mathbf{v} \in \mathbb{R}^n$ is defined

as $\|\mathbf{v}\|_0 = \sum_{i=1}^{n} |v_i|^0$. The Frobenius norm of the matrix $\mathbf{M} \in \mathbb{R}^{s\times t}$ is defined as $\|\mathbf{M}\|_F = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{m} m_{ij}^2}$.

For the matrix $\mathbf{M}$, its $\ell_{2,1}$-norm is defined as

$$\|\mathbf{M}\|_{2,1} = \sum_{i=1}^{n}\sqrt{\sum_{j=1}^{m} m_{ij}^2} = \sum_{i=1}^{n}\|\mathbf{m}^i\|_2. \tag{1}$$

The $\ell_{2,1}$-norm can be generalized to $\ell_{r,p}$-norm

$$\|\mathbf{M}\|_{r,p} = \left(\sum_{i=1}^{n}\left(\sum_{j=1}^{m}|m_{ij}|^r\right)^{\frac{p}{r}}\right)^{\frac{1}{p}} = \left(\sum_{i=1}^{n}\|\mathbf{m}^i\|_r^p\right)^{\frac{1}{p}}. \tag{2}$$

When $r \geq 1$ *and* $p \geq 1$, the $\ell_{r,p}$-norm is valid because it satisfies the three norm conditions including the triangle inequality $\|\mathbf{A}\|_{r,p} + \|\mathbf{B}\|_{r,p} \geq \|\mathbf{A}+\mathbf{B}\|_{r,p}$. However, when $0 \leq r < 1$ *or* $0 \leq p < 1$, $\ell_{r,p}$-norm is not valid as well as the ℓ_0-norm, but we also call them norms for convenience.

In this paper, we denote $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\} \in \mathbb{R}^{d\times n}$ as the training data set, where d is the dimension of features and n is the total number of training instances. The training data $\mathbf{X}$ is labeled by its corresponding labels $\mathbf{a} = \{a_1, a_2, \ldots, a_n\}$, where the label of the ith sample $\mathbf{x}_i$ is a_i. In order to reduce the influence from label data to training data set, here we use $\mathbf{Y} = \{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n\} \in \{0,1\}^{c\times n}$ as the label where the a_ith term of $\mathbf{y}_i$ equals to 1 and the other terms equal to 0. In the minimization problem, we consider learning a projection matrix $\mathbf{W} \in \mathbb{R}^{d\times c}$ which help us to select features.

2.2. $\ell_{2,1}$-norm based feature selection

RFS [13] is an efficient and robust feature selection method via joint $\ell_{2,1}$-norms minimization, which is desired to extract meaningful features and eliminate noisy ones. RFS emphasizes joint $\ell_{2,1}$-norm minimization on both loss function and regularization. It is applied into both genomic and proteomic biomarkers discovery. And an efficient algorithm is given to solve the minimization problem.

The RFS method optimizes

$$\min_{\mathbf{W}} J(\mathbf{W}) = \|\mathbf{W}^T\mathbf{X} - \mathbf{Y}\|_{2,1} + \gamma\|\mathbf{W}\|_{2,1}. \tag{3}$$

Afterwards, inspired by the $\ell_{2,1}$-norm based multi-task learning and feature selection, Metsis et al. propose a newly developed feature selection method based on structured sparse regularization that produces higher accuracy tested on aCGH data [45].

2.3. Capped-ℓ_1, ℓ_1 regularization

In this section, we introduce a non-convex formulation for multi-task feature learning, based on the capped-ℓ_1, ℓ_1 regularization. It considers learning a weight matrix $\mathbf{W} \in \mathbb{R}^{d\times c}$ consisting of the weight vectors for n linear predictive models: $\mathbf{y}_i \approx \mathbf{f}_i(\mathbf{X}_i) = \mathbf{X}_i \times \mathbf{w}_i$. Specifically they impose the ℓ_1 penalty on each row of $\mathbf{W}$, obtaining a column vector. Then they impose the capped-ℓ_1 penalty on that vector and formulate the model as follows:

$$capped-\ell_1, \ell_1 : \min_{\mathbf{W}}\left\{l(\mathbf{W}) + \lambda\sum_{j=1}^{d}\min(\|\mathbf{w}^j\|_1, \varepsilon)\right\}, \tag{4}$$

where $l(\mathbf{W})$ is an empirical loss function [46].

3. Feature selection via simultaneous capped ℓ_2-norm and $\ell_{2,p}$-norm minimization

In this section, we will introduce the formulation of our method and then propose a program to solve its objective function.

3.1. Formulation of SCM

In statistics, least squares is a method for estimating the unknown parameters in a linear regression model [47], with the goal of minimizing the differences between the observed responses in some arbitrary data sets and the responses predicted by the linear approximation of the data. Least squares regression is applied in many tasks for classification, and it is also a popular method for feature selection. In this paper, we use traditional least squares regression to build our formulation.

Most data sets given from the internet include the training data $\mathbf{X}$ and the associated label $\mathbf{a}$. Here we want to structure a traditional least squares regression and the biggest issue is that training data $\mathbf{X}$ is a matrix and the label $\mathbf{a}$ is a vector. Besides, using $\mathbf{Y}$ as the label data is a common strategy to deal with multi-class classification, which converts multi-class classification to some binary classification tasks via one-vs-rest strategy. So we use $\mathbf{Y}$ as the labeled data instead of $\mathbf{a}$. Besides, $\mathbf{Y}$ is more applicable to feature selection because the number of label will not impact the result any more.

With training data $\mathbf{X}$ and the associated class labels $\mathbf{Y}$, traditional least squares regression solves the following formulation to obtain the projection matrix $\mathbf{W}$:

$$\min_{\mathbf{W},\mathbf{b}}\sum_{i=1}^{n}\|\mathbf{W}^T\mathbf{x}_i + \mathbf{b} - \mathbf{y}_i\|_2^2, \tag{5}$$

where $\mathbf{b} \in \mathbb{R}^c$ is the bias. We can absorb it into $\mathbf{W}$ when the constant value 1 is added as an additional dimension for each data $\mathbf{x}_i$. Then the problem becomes:

$$\min_{\mathbf{W}}\sum_{i=1}^{n}\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2^2. \tag{6}$$

The residual $\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2^2$ is squared and thus outliers have more importance than the non squared one. Meanwhile, the pure ℓ_1-norm loss function $\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|$ has less desirable property than ℓ_2-norm [48]. So we choose $\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2$ to be the residual ultimately.

However, ℓ_2-norm loss function is also susceptible to the outliers even if better than the squared one in training data. The $\ell_{2,1}$-norm loss is more robust to the noise, but it requires a large amount of labeled samples and does not consider the influence of outliers. In real world applications, the outliers often appear in the data sets and the labeled data is always few while there are large amount of unlabeled samples existing. Therefore, the traditional loss functions may not get the optimal performance.

It is obvious that the best way to deal with outliers is to detect and reduce them. So we add a capped norm to our ℓ_2-norm loss function to improve the robustness of the regression model [41]:

$$\min_{\mathbf{W}}\sum_{i=1}^{n}\min(\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2, \varepsilon). \tag{7}$$

where ε is the thresholding parameter.

If the kth element of $\mathbf{X}$ is an outlier, $\mathbf{x}_k$ will be far apart from the regression. Then the value of $\|\mathbf{W}^T\mathbf{x}_k - \mathbf{y}_k\|_2$ will be very large. As a result, we use the capped norm and the parameter ε to cut down the impact of outliers. In this case, the analysis of the thresholding parameter ε will be of great significance. After rigorous analysis and experiments, we reach a conclusion that the parameter ε takes influence on the final results and our method is robust when it changes within a certain range.

Then, the next step we should do is to add a regularization term $\mathbf{R}(\mathbf{W})$ with parameter γ to prevent overfitting and to improve interpretability of the learned model. Different regression models have various loss functions and regularizers. In literature,

there are many ways to define the regularization: $\mathbf{R}_1 = \|\mathbf{W}^2\|$ is the ridge regularization. $\mathbf{R}_2 = \sum_{j=1}^{c} \|\mathbf{w}_j\|_1$ is the LASSO regularization. $\mathbf{R}_3 = \sum_{i=1}^{d} \|\mathbf{w}^i\|_2^0$ and $\mathbf{R}_4 = (\sum_{i=1}^{d} \|\mathbf{w}^i\|_2^p)^{\frac{1}{p}}$ penalizes all regression coefficients corresponding to a single feature as a whole with $0 < p \le 1$. This has effects on feature selection [13]. From the sparsity perspective, the ℓ_0-norm of $\mathbf{R}_3$ will be the ideal one [49]. However $\ell_{2,0}$-norm minimization problem is a NP-hard combinational optimization problem and can be very difficult to solve directly, due to its non-convex and non-smooth properties for optimization. Moreover, as defined in Eq. (2), p is set to add sparse constraint for feature selection and r is set to measure the norm of each row vector. The reason why we choose $r = 2$ is for solution and when $r > 2$, it will not take large influence on the final results. Therefore, a lot of researchers prefer to choose $\mathbf{R}_4$ because the $\ell_{2,p}$-norm minimization has rotational invariant property.

In this paper, we use the p power of $\ell_{2,p}$-norm to measure its contribution in regression. The reason is that the regularization $\|\mathbf{W}\|_{2,p}^p$ can achieve the same results as $\|\mathbf{W}\|_{2,p}$ with the regulation of γ and we can get a more efficient approach to solve the formulation with the p power.

In conclusion, we propose the formulation of our method as follows:

$$\min_{\mathbf{W}} \sum_{i=1}^{n} \min(\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2, \varepsilon) + \gamma \|\mathbf{W}\|_{2,p}^p \tag{8}$$

3.2. Optimization and solution

In this section, we will propose an efficient method to solve the formulation.

Firstly, we denote diagonal matrix $\mathbf{F} \in \mathbb{R}^{n\times n}$ and the ith element of $\mathbf{F}$ is:

$$\mathbf{F}_{ii} = \frac{1}{2}\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2^{-1} I(\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2 \le \varepsilon), \tag{9}$$

where $I(\cdot)$ is an indicative function, which is equal to 1 if $\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2 \le \varepsilon$ and 0 otherwise. Then the formulation in Eq. (8) will be converted into the following problem:

$$\min_{\mathbf{W}} \sum_{i=1}^{n} F_{ii}\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2^2 + \gamma \|\mathbf{W}\|_{2,p}^p. \tag{10}$$

As $\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2^2$ is equivalent to $(\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i)^T(\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i)$. We can expand Eq. (10) to the following formulation:

$$\min_{\mathbf{W}} \sum_{i=1}^{n} F_{ii}(\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i)^T(\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i) + \gamma \|\mathbf{W}\|_{2,p}^p. \tag{11}$$

According to linear algebra, we can reformulate Eq. (11) in matrix form as follows:

$$J(\mathbf{W}, \mathbf{F}) = Tr[(\mathbf{W}^T\mathbf{X} - \mathbf{Y})\mathbf{F}(\mathbf{W}^T\mathbf{X} - \mathbf{Y})^T] + \gamma \|\mathbf{W}\|_{2,p}^p, \tag{12}$$

Recall the idea in solving the $\ell_{2,1}$ regularization problem as in [13]. We take the derivative of the regularization $\|\mathbf{W}\|_{2,p}$ with respect to $\mathbf{W}$ as follows:

$$\frac{\partial \|\mathbf{W}\|_{2,p}^p}{\partial \mathbf{W}} = \mathbf{D}\mathbf{W}, \tag{13}$$

where $\mathbf{D} \in \mathbb{R}^{d\times d}$ is a diagonal matrix with the ith element as:

$$\mathbf{D}_{ii} = \frac{p}{2}\|\mathbf{w}^i\|_2^{p-2}. \tag{14}$$

Recalling the definition of $\mathbf{D}_{ii}$ in Eq. (14), if $\mathbf{w}^i \neq 0$, $Tr(\mathbf{W}^T\mathbf{D}\mathbf{W}) = \|\mathbf{W}\|_{2,p}^p/2$. So the minimization of $Tr(\mathbf{W}^T\mathbf{D}\mathbf{W})$ will add the row sparsity on $\mathbf{W}$ [13]. Then the derivative of J in Eq. (12) can be regarded as the derivative of the following objective function:

$$J(\mathbf{W}, \mathbf{F}, \mathbf{D}) = Tr[(\mathbf{W}^T\mathbf{X} - \mathbf{Y})\mathbf{F}(\mathbf{W}^T\mathbf{X} - \mathbf{Y})^T] + \gamma Tr(\mathbf{W}^T\mathbf{D}\mathbf{W}). \tag{15}$$

Now, we adopt the objective function in Eq. (15) to approximate the SCM formulation in Eq. (8). Since $\mathbf{F}$ and $\mathbf{D}$ are dependant on $\mathbf{W}$, Eq. (15) is a regularized least square problem when $\mathbf{F}$ and $\mathbf{D}$ are fixed. Thus, we take derivative of $J(\mathbf{W}, \mathbf{F}, \mathbf{D})$ with respect to $\mathbf{W}$ and set it to zero, and then we can arrive at the update formulation of $\mathbf{W}$ as follows:

$$\frac{\partial J(\mathbf{W}, \mathbf{F}, \mathbf{D})}{\partial \mathbf{W}} = \gamma\mathbf{D}\mathbf{W} + \mathbf{X}\mathbf{F}\mathbf{X}^T\mathbf{W} - \mathbf{X}\mathbf{F}\mathbf{Y}^T = 0. \tag{16}$$

$$\mathbf{W} = (\mathbf{X}\mathbf{F}\mathbf{X}^T + \gamma\mathbf{D})^{-1}\mathbf{X}\mathbf{F}\mathbf{Y}^T. \tag{17}$$

We can update $\mathbf{F}$ and $\mathbf{D}$ by employing the formulation Eq. (9) and Eq. (14), respectively, with $\mathbf{W}$ computed. $\mathbf{W}$ is a optimum solution to the problem if the Eq. (17) is satisfied. In conclusion, we propose an iterative algorithm in this paper to obtain the solution $\mathbf{W}$ so that Eq. (8) is satisfied.

In each iteration, $\mathbf{W}$ is calculated with the current $\mathbf{F}$ and $\mathbf{D}$, and then $\mathbf{F}$ and $\mathbf{D}$ are updated by the current calculated $\mathbf{W}$. We repeated the iteration procedure until the algorithm converges. In addition, the experiment about the convergence behaviour of our method shows that our proposed algorithm converges fast. The times of iterations is often less than 10. The main steps of our method is described in Algorithm 1.

Algorithm 1 SCM.

Data : $\mathbf{X} \in \mathbb{R}^{d\times n}$, $\mathbf{Y} \in \mathbb{R}^{c\times n}$.
Result : $\mathbf{W} \in \mathbb{R}^{d\times c}$
Set $t = 0$, Initialize $\mathbf{F}_t \in \mathbb{R}^{n\times n}$, $\mathbf{D}_t \in \mathbb{R}^{d\times d}$ as identity matrices
repeat
  Calculate $\mathbf{W}_t = (\mathbf{X}\mathbf{F}_t\mathbf{X}^T + \gamma\mathbf{D}_t)^{-1}\mathbf{X}\mathbf{F}_t\mathbf{Y}^T$
  Calculate the diagonal matrices $\mathbf{F}_{t+1}$ and $\mathbf{D}_{t+1}$, where
  $\mathbf{F}_{ii} = \frac{1}{2}\|\mathbf{W}_t^T\mathbf{x}_i - \mathbf{y}_i\|_2^{-1} I(\|\mathbf{W}_t^T\mathbf{x}_i - \mathbf{y}_i\|_2 \le \varepsilon)$
  $\mathbf{D}_{ii} = \frac{p}{2}\|\mathbf{w}_t^i\|_2^{p-2}$
  $t = t + 1$
until Converges

Remark 1. When we compute $\mathbf{D}$, its diagonal element $\mathbf{D}_{ii} = \frac{p}{2}\|\mathbf{w}^i\|_2^{p-2}$. In practice $\|\mathbf{w}^i\|_2$ could be very close to zero but not zero. In theory, it can be zero nonetheless. Therefore, we regularize $\mathbf{D}_{ii} = \frac{p}{2}((\mathbf{w}^i)^T\mathbf{w}^i + \mu)^{(p-2)/2}$, where μ is a very small constant. When $\mu \to 0$, it is easy to see that the regularized $\mathbf{D}_{ii}$ approximates $\frac{p}{2}\|\mathbf{w}^i\|_2^{p-2}$.

4. Algorithm analysis

4.1. Convergence analysis

In this section, we provide a theoretical analysis that our iterative method will monotonically decrease the objective function value of Eq. (8) in each iteration process. We will propose two propositions to show that we can obtain the optimal solution in each step.

Lemma 1. *For any nonzero vectors* $\mathbf{v}$ *and* $\mathbf{v}_k$, *the following inequality holds:*

$$\frac{\|\mathbf{v}\|_2}{\|\mathbf{v}_k\|_2} - \frac{1}{2}\frac{\|\mathbf{v}\|_2^2}{\|\mathbf{v}_k\|_2^2} \le \frac{1}{2}. \tag{18}$$

This lemma was proposed by Nie et al. [13], and proved particularly in [13] as well.

Proposition 1. *The procedures of SCM shown in Algorithm 1 will monotonically decrease the objective function of Eq. (15) in each step.*

Proof. According to Lemma 1, we have

$$\frac{\|\mathbf{w}^i_{(k+1)}\|_2^p}{\|\mathbf{w}^i_{(k)}\|_2^p} - \frac{p}{2}\frac{\|\mathbf{w}^i_{(k+1)}\|_2^2}{\|\mathbf{w}^i_{(k)}\|_2^2} \le 1 - \frac{p}{2}. \tag{19}$$

The above equation is equivalent to the following inequality:

$$\|\mathbf{w}^i_{(k+1)}\|_2^p - \frac{p}{2}\frac{\|\mathbf{w}^i_{(k+1)}\|_2^2}{\|\mathbf{w}^i_{(k)}\|_2^{2-p}} \le \|\mathbf{w}^i_{(k)}\|_2^p - \frac{p}{2}\frac{\|\mathbf{w}^i_{(k)}\|_2^2}{\|\mathbf{w}^i_{(k)}\|_2^{2-p}}. \tag{20}$$

Thus, the following inequality holds:

$$\sum_{i=1}^{d}\left(\|\mathbf{w}^i_{(k+1)}\|_2^p - \frac{p}{2}\frac{\|\mathbf{w}^i_{(k+1)}\|_2^2}{\|\mathbf{w}^i_{(k)}\|_2^{2-p}}\right) \le \sum_{i=1}^{d}\left(\|\mathbf{w}^i_{(k)}\|_2^p - \frac{p}{2}\frac{\|\mathbf{w}^i_{(k)}\|_2^2}{\|\mathbf{w}^i_{(k)}\|_2^{2-p}}\right). \tag{21}$$

Assume that we have derived $\mathbf{D}$ and $\mathbf{F}$ as $\mathbf{D}_{(k)}$ and $\mathbf{F}_{(k)}$ in the kth step. And in the $(k+1)$th iteration, we fix $\mathbf{D}$ and $\mathbf{F}$ as $\mathbf{D}_{(k)}$ and $\mathbf{F}_{(k)}$ and then optimize $\mathbf{W}$.

Then the optimization problem can be reformulated as:

$$(\mathbf{W}_{(k+1)}) = \arg\min_{\mathbf{W}} J(\mathbf{W}, \mathbf{F}_{(k)}, \mathbf{D}_{(k)}). \tag{22}$$

When $\mathbf{F}$ and $\mathbf{D}$ are fixed in Eq. (15), the formulation in Eq. (15) is the regularized least square regression model. In other word, there is a closed solution for Eq. (15) in this case. So we have the following inequality:

$$J(\mathbf{W}_{(k+1)}, \mathbf{F}_{(k)}, \mathbf{D}_{(k)}) \le J(\mathbf{W}_{(k)}, \mathbf{F}_{(k)}, \mathbf{D}_{(k)}), \tag{23}$$

which presents that

$$\begin{aligned} &Tr[(\mathbf{W}^T_{(k+1)}\mathbf{X} - \mathbf{Y})\mathbf{F}_{(k)}(\mathbf{W}^T_{(k+1)}\mathbf{X} - \mathbf{Y})^T] + \gamma Tr(\mathbf{W}^T_{(k+1)}\mathbf{D}_{(k)}\mathbf{W}_{(k+1)}) \\ &\le Tr[(\mathbf{W}^T_{(k)}\mathbf{X} - \mathbf{Y})\mathbf{F}_{(k)}(\mathbf{W}^T_{(k)}\mathbf{X} - \mathbf{Y})^T] + \gamma Tr(\mathbf{W}^T_{(k)}\mathbf{D}_{(k)}\mathbf{W}_{(k)}). \end{aligned} \tag{24}$$

That is to say,

$$\begin{aligned} &Tr[(\mathbf{W}^T_{(k+1)}\mathbf{X} - \mathbf{Y})\mathbf{F}_{(k)}(\mathbf{W}^T_{(k+1)}\mathbf{X} - \mathbf{Y})^T] + \gamma\left(\sum_{i=1}^{d}\frac{p}{2}\frac{\|\mathbf{w}^i_{(k+1)}\|_2^2}{\|\mathbf{w}^i_{(k)}\|_2^{2-p}}\right) \\ &\le Tr[(\mathbf{W}^T_{(k)}\mathbf{X} - \mathbf{Y})\mathbf{F}_{(k)}(\mathbf{W}^T_{(k)}\mathbf{X} - \mathbf{Y})^T] + \gamma\left(\sum_{i=1}^{d}\frac{p}{2}\frac{\|\mathbf{w}^i_{(k)}\|_2^2}{\|\mathbf{w}^i_{(k)}\|_2^{2-p}}\right), \end{aligned} \tag{25}$$

where $\mathbf{w}^i_{(k+1)}$ and $\mathbf{w}^i_{(k)}$ are the ith row of the matrix $\mathbf{W}_{(k+1)}$ and $\mathbf{W}_{(k)}$, respectively. Combining Eqs. (21) and(25), we can get

$$\begin{aligned} &Tr[(\mathbf{W}^T_{(k+1)}\mathbf{X} - \mathbf{Y})\mathbf{F}_{(k)}(\mathbf{W}^T_{(k+1)}\mathbf{X} - \mathbf{Y})^T] + \gamma\sum_{i=1}^{d}\|\mathbf{w}^i_{(k+1)}\|_2^p \\ &\le Tr[(\mathbf{W}^T_{(k)}\mathbf{X} - \mathbf{Y})\mathbf{F}_{(k)}(\mathbf{W}^T_{(k)}\mathbf{X} - \mathbf{Y})^T] + \gamma\sum_{i=1}^{d}\|\mathbf{w}^i_{(k)}\|_2^p. \end{aligned} \tag{26}$$

Then, the equation is equivalent to the following inequality:

$$J(\mathbf{W}_{(k+1)}, \mathbf{F}_{(k)}, \mathbf{D}_{(k+1)}) \le J(\mathbf{W}_{(k)}, \mathbf{F}_{(k)}, \mathbf{D}_{(k)}). \tag{27}$$

And then we will update $\mathbf{F}_{(k)}$ with $\mathbf{W}_{(k+1)}$. The next step is to prove that the procedure of updating $\mathbf{F}_{(k)}$ also decreases the objective function value. In other words, we want to proof the inequality as follows:

$$J(\mathbf{W}_{(k+1)}, \mathbf{F}_{(k+1)}, \mathbf{D}_{(k+1)}) \le J(\mathbf{W}_{(k+1)}, \mathbf{F}_{(k)}, \mathbf{D}_{(k+1)}). \tag{28}$$

Similar with [46], we define two auxiliary functions:

$$\begin{aligned} &\mathbf{h} : \mathbb{R}^{c\times n} \longmapsto \mathbb{R}^n_+, \mathbf{h}(\mathbf{A}) = [\|\mathbf{A}_1\|_2, \ldots, \|\mathbf{A}_n\|_2]^T \\ &\mathbf{g} : \mathbb{R}^n_+ \longmapsto \mathbb{R}_+, \mathbf{g}(\mathbf{u}) = \sum_{i=1}^{n}\min(\mathbf{u}_i, \varepsilon) \end{aligned} \tag{29}$$

Note that $\mathbf{g}(\cdot)$ is a concave function and we say that a vector $\mathbf{s} \in \mathbb{R}^n$ is a sub-gradient of $\mathbf{g}$ at $\mathbf{v} \in \mathbb{R}^n_+$, if for all vector $\mathbf{u} \in \mathbb{R}^n_+$, the inequality holds:

$$\mathbf{g}(\mathbf{u}) \le \mathbf{g}(\mathbf{v}) + \langle \mathbf{s}, \mathbf{u} - \mathbf{v}\rangle, \tag{30}$$

where $\langle\cdot,\ \cdot\rangle$ denotes the inner product. Define $\mathbf{A}_i = \mathbf{W}\mathbf{x}_i - \mathbf{y}_i$, $\mathbf{u} = \mathbf{h}(\mathbf{A})$, we can rewrite the original formulation of SCM Eq. (8) as follows:

$$\min_{\mathbf{W}} \mathbf{g}(\mathbf{h}(\mathbf{A})) + \gamma R(\mathbf{W}), R(\mathbf{W}) = Tr(\mathbf{W}^T\mathbf{D}\mathbf{W}) \tag{31}$$

Based on the definition of the sub-gradient for a concave function given above, we can obtain an upper bound of $\mathbf{g}(\mathbf{h}(\mathbf{A}))$ using a locally linear approximation at $\mathbf{h}(\mathbf{A}^{(k)})$:

$$\mathbf{g}(\mathbf{h}(\mathbf{A})) \le \mathbf{g}(\mathbf{h}(\mathbf{A}^{(k)})) + \langle \mathbf{s}^{(k)}, \mathbf{h}(\mathbf{A}) - \mathbf{h}(\mathbf{A}^{(k)})\rangle, \tag{32}$$

where $\mathbf{s}^{(k)}$ is a sub-gradient of $\mathbf{g}(\mathbf{u})$ at $\mathbf{u} = \mathbf{h}(\mathbf{A}^{(k)})$. Furthermore, we can obtain an upper bound of the objective function in Eq. (31), if the solution $\mathbf{A}^{(k)}$ at the kth iteration is available:

$$\mathbf{g}(\mathbf{h}(\mathbf{A})) + \gamma R(\mathbf{W}) \le \mathbf{g}(\mathbf{h}(\mathbf{A}^{(k)})) + \langle \mathbf{s}^{(k)}, \mathbf{h}(\mathbf{A}) - \mathbf{h}(\mathbf{A}^{(k)})\rangle + \gamma R(\mathbf{W}). \tag{33}$$

It can be shown that a sub-gradient of $\mathbf{g}(\mathbf{h}(\mathbf{A}))$ at $\mathbf{h}(\mathbf{A}^{(k)}))$ is:

$$\mathbf{s}^{(k)} = \frac{1}{2}[I(\|\mathbf{A}^{(k)}_1\|_2, \varepsilon), I(\|\mathbf{A}^{(k)}_2\|_2, \varepsilon), \ldots, I(\|\mathbf{A}^{(k)}_n\|_2, \varepsilon)] \tag{34}$$

Since $\mathbf{h}(\mathbf{A}^{(k)})$ and γ are constant with respect to $\mathbf{W}$, we have:

$$\begin{aligned} \mathbf{W}_{(k+1)} &= \arg\min_{\mathbf{W}}[\mathbf{g}(\mathbf{h}(\mathbf{A}^{(k)})) + \langle \mathbf{s}^{(k)}, \mathbf{h}(\mathbf{A}) - \mathbf{h}(\mathbf{A}^{(k)})\rangle] + \gamma R(\mathbf{W}) \\ &= \langle \mathbf{s}^{(k)}, \mathbf{h}(\mathbf{A})\rangle + \gamma R(\mathbf{W}), \end{aligned} \tag{35}$$

which obtains the next iterative solution by minimizing the upper bound of the objective function in Eq. (33). Thus, in the viewpoint of the locally linear approximation, we can understand Algorithm 1 as follows: the original formulation of SCM is non-convex and is difficult to solve; and then the proposed method minimizes an upper bound in each step, which is convex and can be solved efficiently. It is closely related to concave convex procedure (CCCP) [50]. In addition, we can easily verify that the objective function value decreases monotonically as follows:

$$\begin{aligned} \mathbf{g}(\mathbf{h}(\mathbf{A}^{(k+1)})) &+ \gamma R(\mathbf{W}) \le \mathbf{g}(\mathbf{h}(\mathbf{A}^{(k)})) + \langle \mathbf{s}^{(k)}, \mathbf{h}(\mathbf{A}^{(k+1)}) - \mathbf{h}(\mathbf{A}^{(k)})\rangle \\ &+ \gamma R(\mathbf{W}) \\ &\le \mathbf{g}(\mathbf{h}(\mathbf{A}^{(k)})) + \langle \mathbf{s}^{(k)}, \mathbf{h}(\mathbf{A}^{(k)}) - \mathbf{h}(\mathbf{A}^{(k)})\rangle + \gamma R(\mathbf{W}) \\ &\le \mathbf{g}(\mathbf{h}(\mathbf{A}^{(k)})) + \gamma R(\mathbf{W}) \end{aligned} \tag{36}$$

According to the definition of $\mathbf{F}$, we know that $\mathbf{s}^{(k)}$ is equivalent to $\mathbf{F}_{(k)}$. Apparently, the inequality in Eq. (28) holds. Finally, combining Eqs. (27) and(28), we can arrive at:

$$J(\mathbf{W}_{(k+1)}, \mathbf{F}_{(k+1)}, \mathbf{D}_{(k+1)}) \le J(\mathbf{W}_{(k)}, \mathbf{F}_{(k)}, \mathbf{D}_{(k)}). \tag{37}$$

This inequality indicates that the objective function in Eq. (15) will monotonically decrease in each iteration. An import issue we should mention is that a monotonic decrease of the objective function value does not guarantee the convergence of the algorithm, even if the objective function is strictly convex and continuously differentiable. □

Proposition 2. *The procedures of SCM shown in Algorithm 1 will monotonically decrease the objective function of Eq. (15) in each step and then converge to the local optimum of Eq. (8).*

Proof. Based on the conclusion of Proposition 1, we adopt the similar strategy in the conference paper [51] to prove it to converge to the local optimum of the Eq. (8).

In each iteration of SCM in Algorithm 1, we find the optimal solution to the problem Eq. (8). Thus the converged solution of SCM satisfies the KKT condition of problem Eq. (8). Taking the derivative

with regard to $\mathbf{W}$ and setting it to zero, we get the KKT condition of the problem Eq. (8) as Eq. (16) in the main parts of this paper.

According to optimization theory, the Lagrangian function of problem of Eq. (8) is:

$$J(\mathbf{W}) = \sum_{i=1}^{n} \min(\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2, \varepsilon) + \gamma \|\mathbf{W}\|_{2,p}^p. \tag{38}$$

Then we take the derivative of the formulation above with reference to $\mathbf{W}$ and set it to zero, we get the KKT condition of the problem Eq. (8) as follows:

$$\frac{\partial J(\mathbf{W})}{\partial \mathbf{W}} = \sum_{i=1}^{n} \frac{1}{2} \|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2^{-1} \mathbf{x}_i(\mathbf{x}_i^T\mathbf{W} - \mathbf{y}_i^T)$$
$$I(\|\mathbf{W}^T\mathbf{x}_i - \mathbf{y}_i\|_2 \le \varepsilon) + 2\gamma \sum_{i=1}^{d} \frac{p}{2} \|\mathbf{w}^i\|_2^{p-2} \mathbf{w}^i = 0. \tag{39}$$

Using the matrix calculus, we can write the above Eq. (39) as follows:

$$\begin{aligned}\frac{\partial J(\mathbf{W})}{\partial \mathbf{W}} &= \sum_{i=1}^{n} \mathbf{F}_{ii}\mathbf{x}_i(\mathbf{x}_i^T\mathbf{W} - \mathbf{y}_i^T) + \gamma \sum_{i=1}^{d} \mathbf{D}_{ii}\mathbf{w}^i \\ &= \gamma \mathbf{D}\mathbf{W} + \sum_{i=1}^{n} \mathbf{F}_{ii}(\mathbf{x}_i\mathbf{x}_i^T\mathbf{W} - \mathbf{x}_i\mathbf{y}_i^T) \\ &= \gamma \mathbf{D}\mathbf{W} + \mathbf{X}\mathbf{F}\mathbf{X}^T\mathbf{W} - \mathbf{X}\mathbf{F}\mathbf{Y}^T = 0\end{aligned} \tag{40}$$

According to the main procedures of SCM, we can see that Eqs. (40) and(16) are the exact fetch of each other when our algorithm is converged. Therefore, the converged solution of Algorithm 1 satisfies Eq. (40), the KKT conditions of problem Eq. (8). Thus the converged solution of Algorithm 1 is a local minimal solution to the problem Eq. (8). □

4.2. Parameter determination

Parameter determination is still an open problem and there are three important parameters in our SCM algorithm. The first is the balance parameter γ, which controls the trade-off between the loss function and the regularization. The larger γ is, the more attention we pay to the regularization $\|\mathbf{W}\|_{2,p}$. In our methods, we determine it by grid search based on cross validation. We vary this parameter within a large range and then determine the approximate optimal parameter when the classification accuracy reaches the peak. The next parameter is the value of p, which balance the sparsity and convexity of the formulation. After further experiment about the $\ell_{2,p}$-norm regularization, we find that the results will not change too much if we adjust p within the range $0 < p \le 1$. So we set $p = 1$ for simpleness when coming to experiment as in previous work in [41]. But the value of parameter p can effect the performance of our method more precisely as the parameter experiment shows. We can determine the value of parameter p by grid search based on cross validation. The third parameter is the threshold parameter ε. In our method, it controls the influence to outliers, we can reduce outliers by decreasing the parameter ε. We get a mapping from ε to the extent of its influence on our method and then determine the optimal parameter by varying the impact when the SVM classification accuracy reaches the peak. From the parameter experiment below, we draw a conclusion that ε takes influence on the final results, however its influence is not so significant when changing within a certain range.

5. Experiment

There are mainly five kinds of experiments in this section. The first experiment is to provide a toy example. The second group mainly focus on the evaluation of classification accuracy with Support Vector Machine (SVM). To demonstrate whether the outperformance of our method is dominated by following classifier, we provide some classification results with K-nearest neighborhood classifier (KNN). The third group is to employ a different kind of evaluation indicator the redundancy rate. The fourth group is to show some results about convergence behavior. Finally, classification accuracies with different parameters are presented. We would like to introduce data sets and evaluation methods at first.

5.1. Data description and evaluation metric

We give a brief description on all data sets used in our experiments as follows:

Isolet[1] data set is collected from 150 speakers who speak the name of each English letter of the alphabet twice. Thus, there are 52 samples from each speaker. The speakers are grouped into 5 subsets which respectively include 30 similar speakers, and the subsets are named Isolet1, Isolet2, Isolet3, Isolet4, and Isolet5. Thus, we naturally have 5 tasks with each task corresponding to one subset. In this paper, we choose the Isolet5 which has 1559 samples including 617 features and the response is the English letter label (1–26).

USPS[2] data set is handwritten digit database contains 20 objects. We use here a popular subset contains 9298 16×16 handwritten digit images in total, which is then split into 7291 training images and 2007 test images.

Coil20[3] data set is used by many researchers as a bench mark data to verify the validity of their proposed object recognition models. The images of each objects were taken 5° intervals (1440 views in total) as the object is rotated on a turntable and each object has 72 images. The size of each image is 32×32 pixels, with 256 grey levels per pixel. Thus, each image is represented by a 1024-dimensional vector.

Musk[4] data set is to predict whether a new drug molecule will bind to a target protein. However each molecule (bag) can have many different low energy shapes (instances) of which only one can actually bind with the target. There are 2 samples such as positive and negative, where 207 examples in positive and 269 examples in the other including 166 features.

Carcinomas data set composed of total 174 samples in eleven classes, prostate, bladder,ureter,breast, colorectal, gastroesophagus, kidney, liver, ovary, pancreas, lung adenocarcinomas, and lung squamous cell carcinoma, which have 26, 8, 26, 23, 12, 11, 7, 27, 6, 14, 14 samples, respectively. In the original data [52], each sample contains 12,533 genes. In the preprocessed data set [53], there are 174 samples and 9182 genes.

Prostate-GE data set has in total 102 samples in two classes tumor and normal, which have 52 and 50 samples, respectively. The original data set contains 12,600 genes. After the preprocessing by [13], we obtained a data set with 102 samples and 5966 genes.

We select different numbers of features and these data sets are employed to validate our algorithms. Additionally, after some preprocessing, the detailed statistical characters of different data sets are listed in Table 1.

To show the effectiveness of our method, we use two different kinds of evaluation metrics on data sets formulated by the selected features, i.e. accuracy-the classification accuracy achieved by classifier using the selected features; redundancy rate-the redundancy rate contained in the selected features. The redundancy is a popular evaluation metric for feature selection. It measures the qual-

[1] http://archive.ics.uci.edu/ml/machine-learning-databases/isolet/.
[2] http://www-i6.informatik.rwth-aachen.de/keysers/usps.html.
[3] http://www1.cs.columbia.edu/CAVE/research/softlib/coil-20.html.
[4] http://www.cc.gatech.edu/~lsong/code.html.

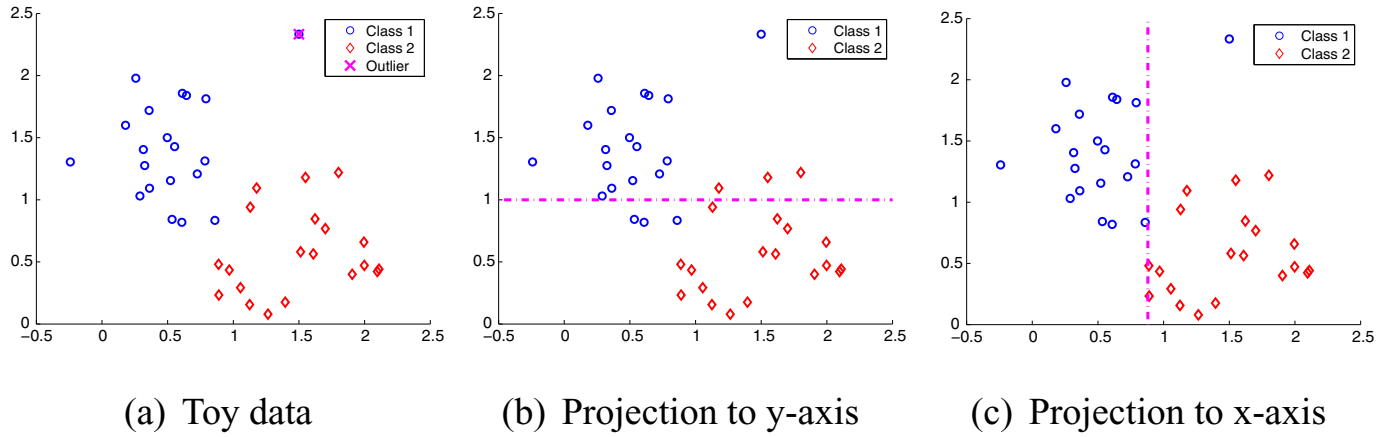


Fig. 2. 2D toy example for feature selection. (a) shows the original data, (b) shows the projection to y-axis (result of RFS), the classification accuracy is 85%, (c) shows the projection to x-axis (result of SCM), the classification accuracy is 97.5%.

Table 1
Characters of different data sets.

Data	Size	Dim	Class	Type	No. of selected fea
Isolet	1559	617	26	Voice	10,15,...,80
USPS	9298	256	20	Image	10,15,...,70
Coil20	1440	1024	20	Image	10,15,...,100
Musk	476	166	2	Biology	20,25,...,80
Carcinomas	174	9182	11	Biology	20,40,...,200
Prostate-GE	102	5966	2	Biology	20,40,...,200

ity of selected features directly, without employing the following tasks. Assume $\mathbf{F}$ is the set of selected features and $\mathbf{X_F}$ is the data represented by the features in $\mathbf{F}$. The following metric is used to measure the redundancy rate of $\mathbf{F}$:

$$\mathrm{RED}(\mathbf{F}) = \frac{1}{|\mathbf{F}|(|\mathbf{F}|-1)} \sum_{\mathbf{f}_i,\mathbf{f}_j \in \mathbf{F}, i>j} \mathrm{corr}_{i,j}, \tag{41}$$

where $|\mathbf{F}|$ is the cardinality of $\mathbf{F}$ and $\mathrm{corr}_{i,j}$ is the correlation between two features f_i and f_j. This measurement assesses the averaged correlation among all feature pairs, and a large value indicates that many selected features are correlated and thus redundancy is expected to exist in $|\mathbf{F}|$.

To compute the classification accuracy, we use the SVM and KNN to perform classification on the data with selected features. We randomly select a fixed number of labeled and unlabeled examples from each category as training data and the rest are assigned as testing data. We perform feature selection on training data and classify all the testing data with the selected features by SVM and KNN. In the following, we compare our method SCM with five other feature selection methods.

5.2. Toy example

We use a 2D toy data experiment to illustrate the robustness of loss function capped ℓ_2-norm comparing with $\ell_{2,p}$-norm. In this experiment, two classes of artificial data samples are generated. We arrange the mean value of the data on x-axis in two classes as 0.5 and 1.5. The data on y-axis of two classes are generated randomly at a similarity of about 50% and then we structure an outlier into one of the two classes as Fig. 2 shows. The two feature selection methods SCM and RFS are presented to apply to the toy example. As a result, RFS project to the y-axis and the classification accuracy is 85%, meanwhile SCM project to the x-axis and the classification accuracy is 97.5%. In addition, when we remove the outlier, RFS will project to x-axis accurately instead. We consider that the reason is that RFS is not robust to outliers but our capped ℓ_2-norm can reduce the influence of outliers effectively. Thus, we can reach a conclusion that SCM is more robust than RFS.

5.3. Classificationaccuracy comparison

In this section, we provide some classification results with different feature selection methods on different data sets. We use Support Vector Machine (SVM) as the classifier. By randomly splitting the original data into training set and testing set for 10 independent runs, we select some samples randomly from each category as training data and the others are set as testing data. With different data sets, the mean classification accuracies of 10 independent runs on all data sets are shown. Then we compare our feature selection method (SCM) to several popularly used feature selection methods, such as T-test, Information gain, ReliefF, Fisher score (Fscore) and Robust Feature Selection (RFS). When proceeding our method, we set the parameter $p = 1$, the value of γ and ε is displayed as follows: Isolet: $\gamma = 0.5$ $\varepsilon = 0.2836$, USPS: $\gamma = 0.1$ $\varepsilon = 0.2239$, Coil20: $\gamma = 0.1$ $\varepsilon = 0.0474$, Musk:$\gamma = 0.1$ $\varepsilon = 0.1399$, Carcinomas: $\gamma = 0.1$ $\varepsilon = 0.0001$, Prostate-GE: $\gamma = 0.1$ $\varepsilon = 0.0003$. Obviously, different data sets have different combinations of optimal parameters.

Fig. 3 shows the classification accuracy comparisons of all six feature selection methods on six data sets. There are three different types of matrix data sets including images data, voice data and biological data. The dimensionality ranges from about 150 to about 9000. We select different numbers of features in line accordance with their dimensionality. Ultimately, we compute the average accuracy using the features for all feature selection approaches.

Different from above settings, we use another classifier KNN for classification. These experiments are proposed to demonstrate that the out-performance of SCM is not dominated by the using of SVM classifier. Considering the data sets we choose, Isolet is voice database, USPS is handwritten digit database, Coil20 is objects image database, Musk, Carcinomas and Prostate-GE are biology databases. Among the biology databases, both the dimensionality and the size of Carcinomas database are the largest. So Carcinomas database can represent biology databases. Finally, we choose four data sets, Isolet, USPS, Coil20 and Carcinomas typically for illustration as Fig. 4 shows.

Obviously our approach outperform other methods significantly as Figs. 3 and4 show, which demonstrates our method performs better in almost all cases. Furthermore, the data sets Carcinomas and Prostate-GE were used in [13] to validate the performance of RFS with the same classifier SVM. According to the performance in

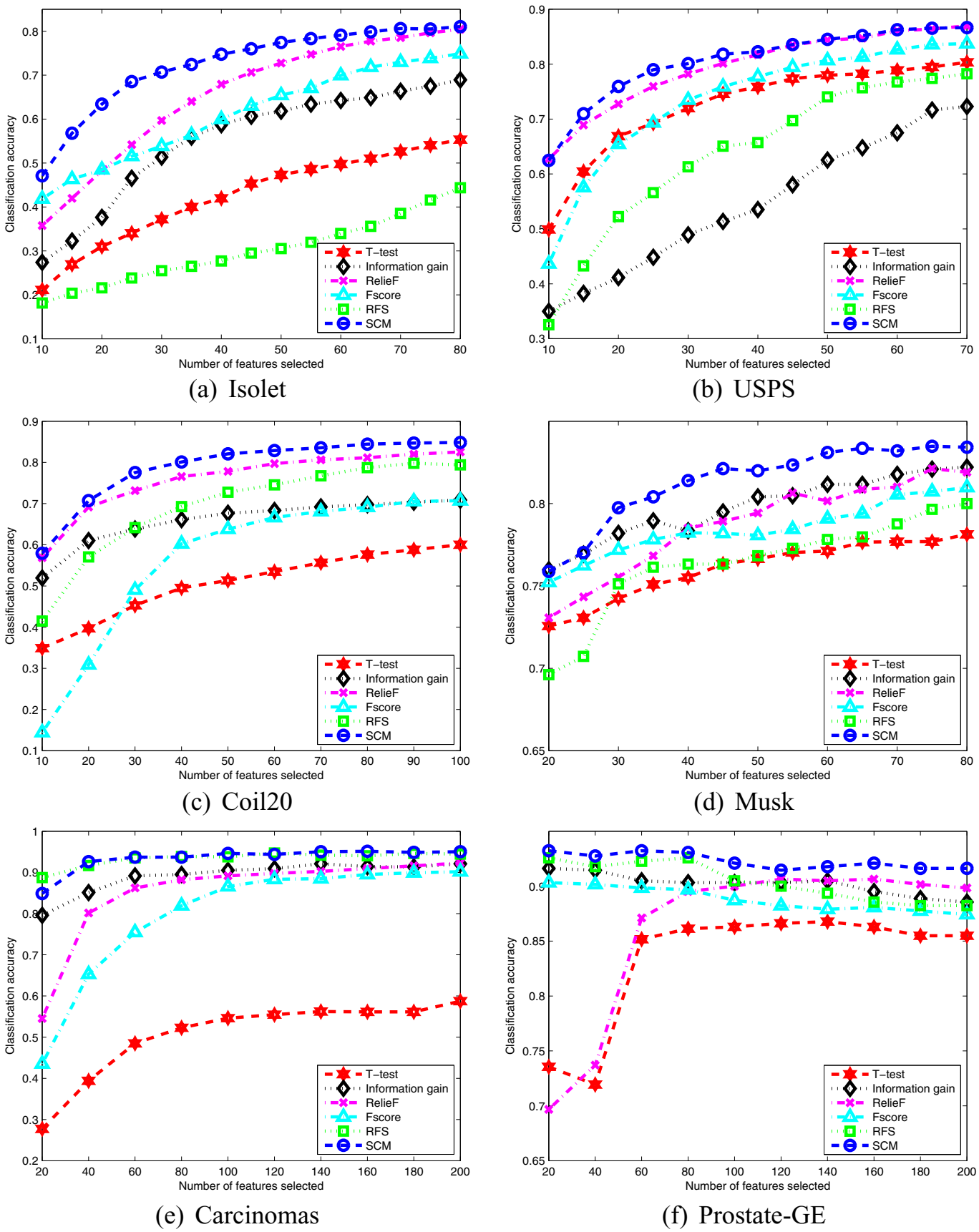


Fig. 3. Classification accuracy comparisons of six feature selection algorithms on six data sets. SVM is used for classification. SCM is our method.

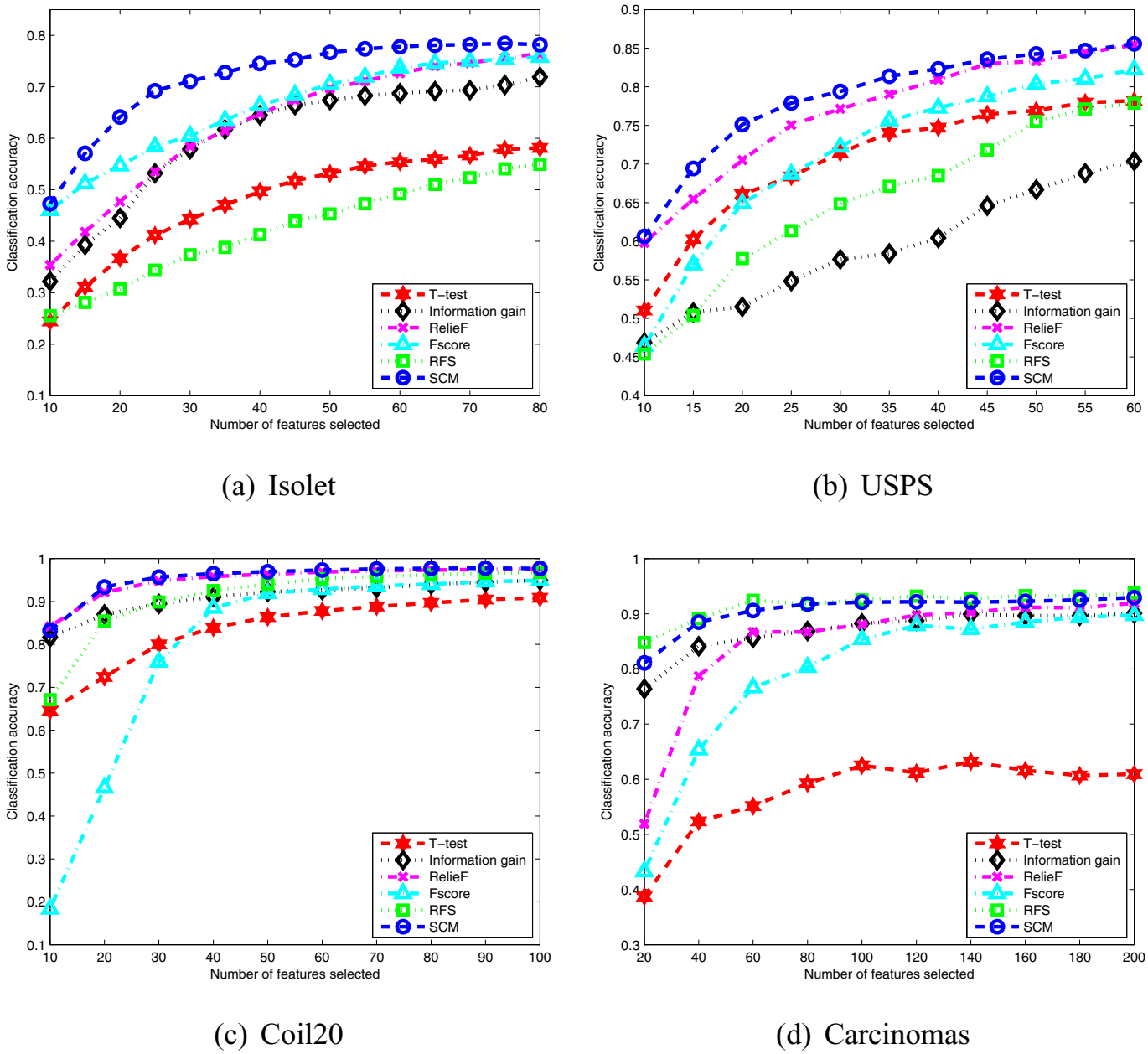


Fig. 4. Classification accuracy comparisons of six feature selection algorithms on four representative data sets. KNN is used for classification. SCM is our method.

Fig. 3, the classification accuracy of our method is also higher than RFS. Meanwhile, it seems that the classification accuracy is rising when the number of selected features is increasing. We consider that it may be because the range we selected from the whole dimension is narrow, when the number of selected features is great in a certain degree, the classification accuracy will show a downward trend. According to the results in Fig. 4, our method performs better as well, this demonstrates that the out-performance of our method is not dominated by the using of SVM classifier. Meanwhile, comparing Figs. 3 and4, we can see that the sort orders of other methods will change when we alter the classifier, but the classification accuracy of our method is always the highest. However, the superiority of our method is not distinct in Carcinomas database with KNN classifier as Fig. 4 shows. We consider that it is because the dimensionality of Carcinomas database is too large and the number of features we selected is a little small.

5.4. Redundancy rate comparison

We compare the redundancy rate (RED) of different methods on emblematic two data sets in this section. We use the same way to get training data and testing data as the classification accuracy comparison. Then we compute the RED using the top 50 features selected by all feature selection approaches.

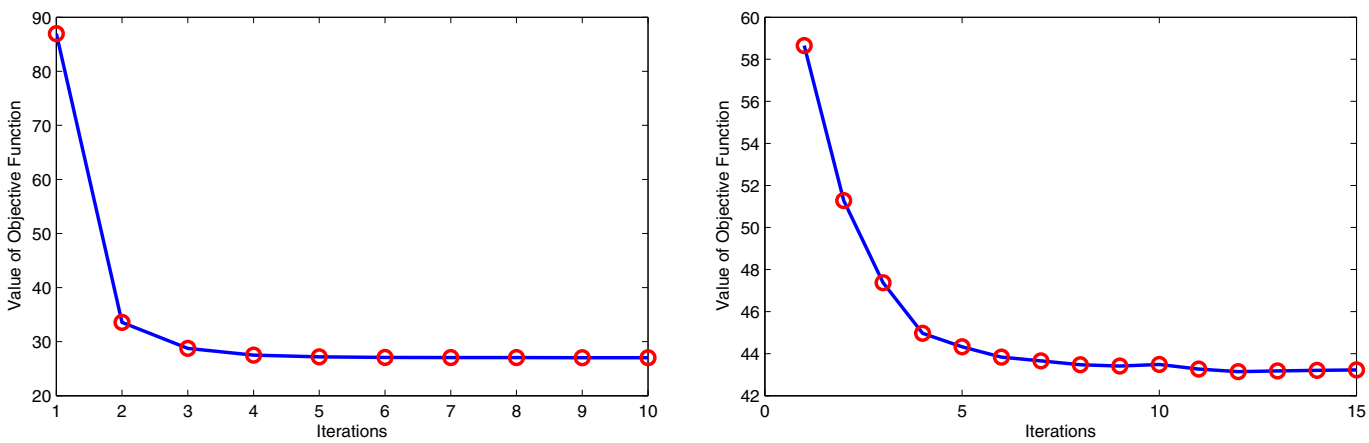
Tables 2–3 present that the feature subsets selected by SCM on data sets have lower redundancy rate than other methods obviously. The RED makes an evaluation directly on the features and does not rely on the classifiers. So the RED comparison result shows that our method is not dominated by classifier, the performance of our method outperform others in many respects. In the mean time, we can see that the RED of our method keeps at a rate that is pretty low from the beginning to the end, this demonstrates that the rank of features our method get is very precise. In addi-

Table 2
The redundancy rate of different feature size selected by different feature selection method on Isolet data. SCM is our method.

No. of selected fea	10	20	30	40	50
T-test	0.6828 ± 0.2668	0.6842 ± 0.1482	0.6918 ± 0.0945	0.6867 ± 0.0702	0.6693 ± 0.0505
Info	0.8741 ± 0.0435	0.8025 ± 0.0364	0.7454 ± 0.0172	0.6920 ± 0.0220	0.6574 ± 0.0214
ReliefF	0.3150 ± 0.0439	0.2897 ± 0.0185	0.2641 ± 0.0162	0.2527 ± 0.0120	0.2445 ± 0.0141
Fscore	0.4299 ± 0.0224	0.5068 ± 0.0218	0.5533 ± 0.0191	0.5612 ± 0.0141	0.5566 ± 0.0086
RFS	0.4007 ± 0.1119	0.3477 ± 0.0651	0.2871 ± 0.0327	0.2597 ± 0.0124	0.2357 ± 0.0082
SCM	$\mathbf{0.2041} \pm 0.0399$	$\mathbf{0.2092} \pm 0.0238$	$\mathbf{0.1982} \pm 0.0224$	$\mathbf{0.2032} \pm 0.0169$	$\mathbf{0.2006} \pm 0.0141$

Table 3
The redundancy rate of different feature size selected by different feature selection method on USPS data. SCM is our method.

No. of selected fea	10	20	30	40	50
T-test	0.5967 ± 0.0797	0.4686 ± 0.0276	0.4267 ± 0.0313	0.3974 ± 0.0278	0.3748 ± 0.0221
Info	0.5973 ± 0.0605	0.5550 ± 0.0335	0.4947 ± 0.0357	0.4654 ± 0.0314	0.4191 ± 0.0173
ReliefF	0.2887 ± 0.0451	0.2555 ± 0.0165	0.2383 ± 0.0155	0.2296 ± 0.0143	0.2209 ± 0.0120
Fscore	0.6610 ± 0.0796	0.4029 ± 0.0512	0.4203 ± 0.0340	0.3667 ± 0.0321	0.3330 ± 0.0247
RFS	0.2654 ± 0.0396	0.2501 ± 0.0381	0.2458 ± 0.0354	0.2434 ± 0.0354	0.2337 ± 0.0262
SCM	$\mathbf{0.2019} \pm 0.0246$	$\mathbf{0.2221} \pm 0.0205$	$\mathbf{0.2167} \pm 0.0229$	$\mathbf{0.2168} \pm 0.0191$	$\mathbf{0.2129} \pm 0.0161$



(a) Isolet (b) USPS

Fig. 5. Convergence behavior of SCM. There are mainly 10 or 15 iterations. The line shows the objective functions value of SCM on different data sets. The *x*-axis represents the number of iterations.

tion, we can see that the RED of other methods will undulate obviously when we change the type of data set, but our method always maintain a low RED level. Hence the generality of our method is better as well.

5.5. Convergence results

In this section, we provide some results to show the convergence behavior of our method in iterative procedure. Data sets Isolet and USPS are employed. We take the objective function as the evaluation criterion and 10 to 15 iterations are shown in Fig. 5.

As seen from the Fig. 5, the objective function of SCM is non-increasing during the iteration. They all converge to a fixed value in the end. Additionally, the convergence is very fast and times of the iteration is less than 10.

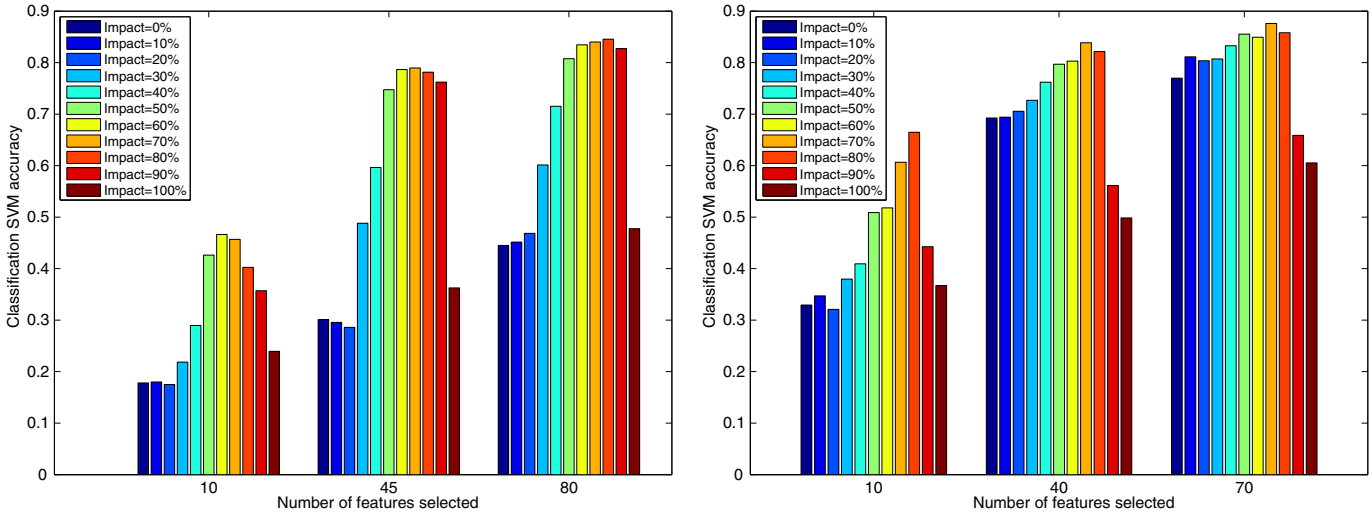
5.6. Classification with different parameters

In this section, we will provide some results with different parameter ε and p.

According to the procedures of SCM shown in algorithm 1, parameter ε affect the result by impacting on the matrix **F**. So we can quantificat the impact of ε to our result by calculating the numbers of elements affected by ε in **F**. Then we can get a mapping from parameter ε to the extent of its influence on our method. As a result we can determine the optimal parameter by varying the impact when the SVM classification accuracy reaches the peak.

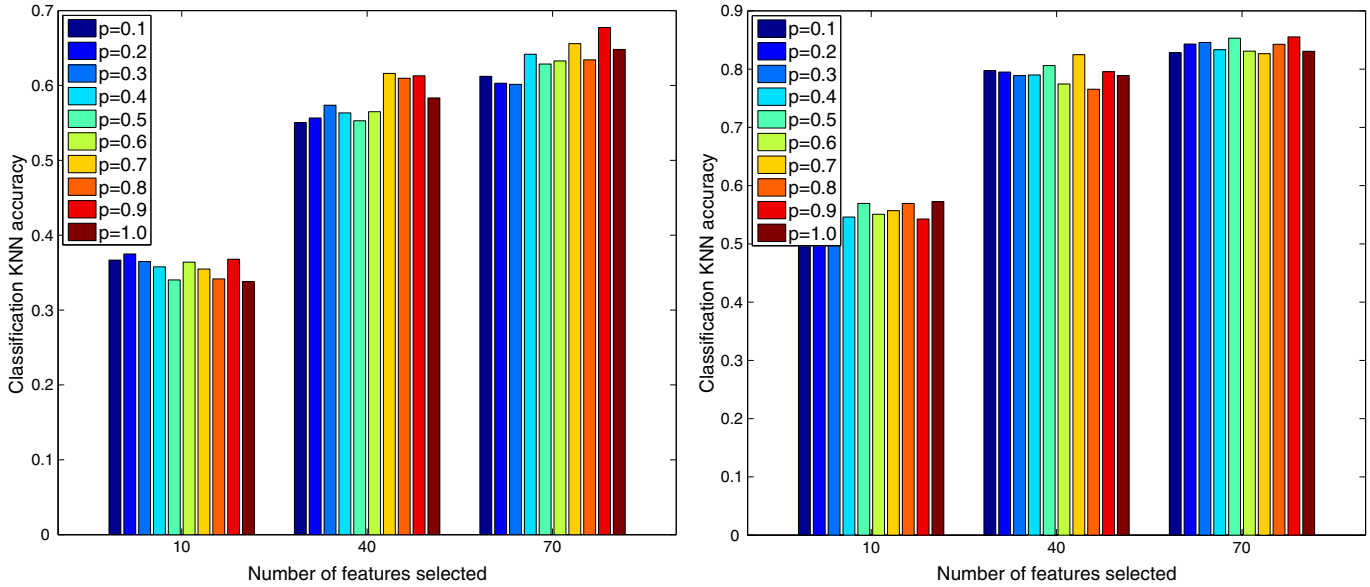
To show the influence of our parameter ε, we employ SVM as the classifier and conduct experiments on two representative data sets Isolet and USPS. We vary the impact of ε as 0%, 10%, 20%,...,100% and deliberate how the classification accuracy changes. With different numbers of training points, we report the mean accuracy of classification in Fig. 6.

As seen from Fig. 6, we can perceive that when the impact comes to 0%, the classification accuracy in Isolet data set is much lower than which in USPS. In fact, we attempt a number of data sets and the outcomes are all on the verge of USPS. We believe that it is because RFS doesn't perform well in Isolet as Fig. 3 shows and SCM verges on RFS when the impact comes to 0%, in other words the loss function turns into ℓ_2-norm. When the impact of parameter gradually increases, the classification accuracy rises as well and then comes to the peak. It is because our parameter is removing outliers and noise while impact rising. When the parameter starts to influence the normal figures as a high degree of affects, the clas-



(a) Isolet (b) USPS

Fig. 6. Classification results of SCM with different parameters ε. Impact shows the extent of parameter's influence on our method.



(a) Isolet (b) USPS

Fig. 7. Classification results of SCM with different parameters p.

sification accuracy begins to descend rapidly. In summary, we can conclude that our parameter takes influence on the final results. Nevertheless, its influence is not so significant if it changes within a certain range.

The parameter p balance the sparsity and convexity of the formulation. To show the influence of our parameter p, we employ KNN as the classifier and conduct experiments on two representative data sets Isolet and USPS. We vary the value of parameter p from 0.1 to 1.0, and then research the changes of the performance of our method.

As seen from Fig. 7, the classification accuracy fluctuates within certain limits when the value of parameter p varies from 0.1 to 1.0. The fluctuation is random and slight so that we can carry out fine tune to the performance of our method by varying p.

6. Conclusion

In this paper, we proposed a new robust and efficient feature selection method with emphasizing simultaneous capped ℓ_2-norm loss and $\ell_{2,p}$-norm regularizer minimization, which is called as SCM. The capped ℓ_2-norm based regression loss function is robust to outliers in data sets and the $\ell_{2,p}$-norm regularization is used to select features across data sets with joint sparsity when $0 < p \leq 1$. An efficient approach was taken to solve the problem. The convergence analysis and result showed the convergence behavior of our method. A 2D toy example was proposed to illustrate the robustness of capped ℓ_2-norm. We used SVM and KNN classification accuracy and redundancy rate to verify the performance of our method comparing with five popular feature selection methods. Fi-

nally, we researched the parameter behaviour concluding that our parameter takes influence on the final results and our method is robust when the parameter changes within a certain range. In the future, we can consider adding capped $\ell_{2,p}$-norm minimization to the loss term. Furthermore, we only focused on the capped norm in feature selection. How to use it in other tasks, such as clustering, ranking etc. is an interesting future work. Besides, how to apply this method in other real applications is also worthwhile to investigate.

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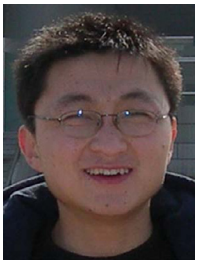

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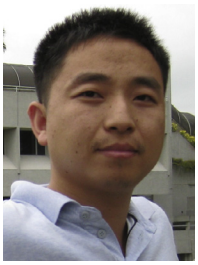

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